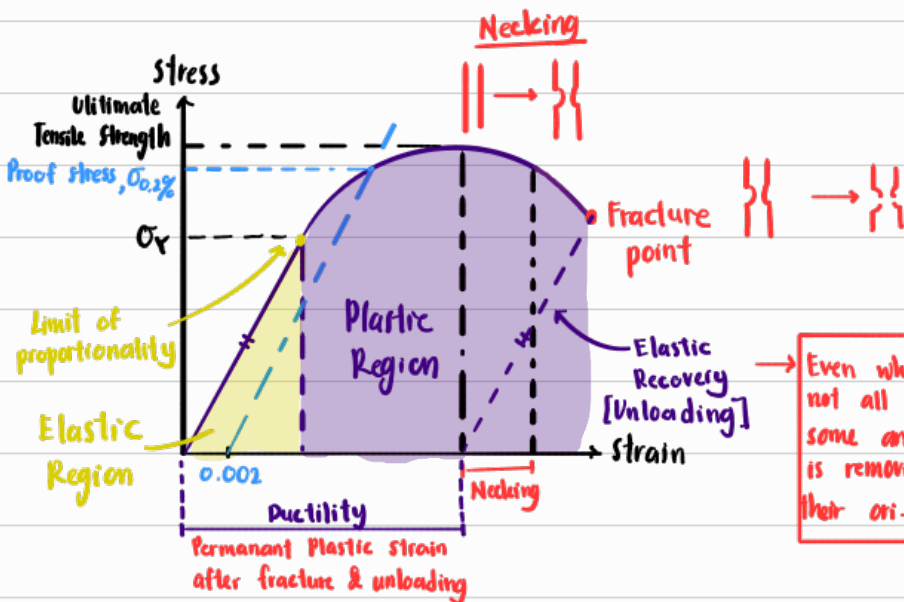


Stress and strain

Stress, $\sigma = \frac{F}{A}$ Amount of force exerted per meter²

Strain, $\epsilon = \frac{\Delta l}{l}$ Degree of extension relative to its original length

Young Modulus, $E = \frac{\sigma}{\epsilon} = \frac{Fl}{A\Delta l}$ Ratio of stress & strain [Stiffness]



Yield stress, σ_y : Stress which permanent deformation

Proof stress, $\sigma_{0.2\%}$: When σ_y can't be determined, we use $\sigma_{0.2\%}$ as an approx. for σ_y

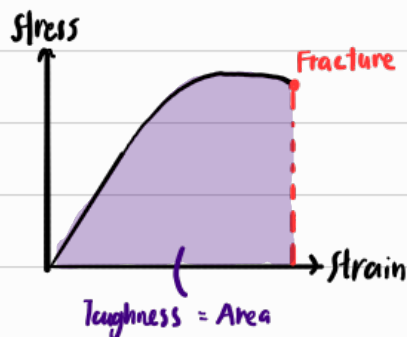
Toughness J/m^3

Work done per unit volume in breaking the material

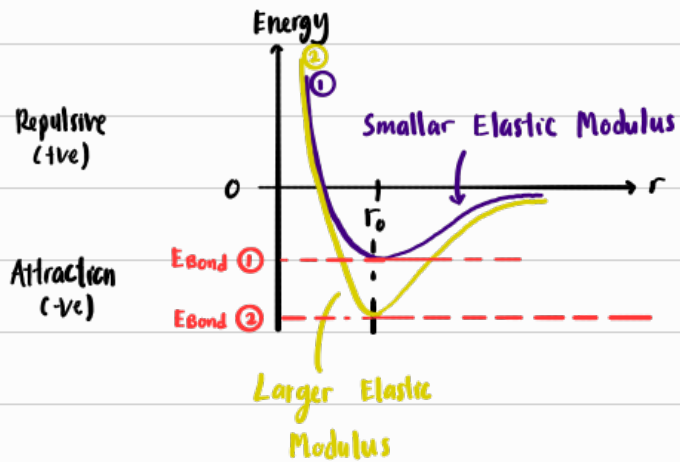
Brittle



Tough



Bonding & Young Modulus



* Bond energy $\uparrow$
Bond strength $\uparrow$
Elastic Modulus $\uparrow$
Melting point $\uparrow$